



Choosing Your First NaNDA Dataset

Breakout 1 · Beginner session · 45 minutes

GSA Annual Scientific Meeting Online Workshop Series 2026 · October 7, 2026

Three questions, one dataset

Where?

Geography

When?

Years

What?

Topics and variables

Your turn

Your own project

HOW IT WORKS

- [ICPSR Account](#)
- [Choosing Your First NaNDA Dataset: Participant Handout \(GSA 2026\)](#)
- Polls, a scavenger hunt, and lots of chat.
- Bring your own project, or borrow Jordan's.



Meet Jordan Ellis



Dr. Jordan Ellis

they|them

Topic	Loneliness among adults 65 and older
Data	100 participants Yearly interviews 2016 to 2019
Geography	Home addresses collected at enrollment in 2016 Geocoded to census tracts

RESEARCH QUESTION

Do older adults living near more places to gather (clubs, community groups, places of worship) report less loneliness?

SEGMENT 1

Where?

Census tracts, ZCTAs, and geographic boundaries, oh my!



Census tracts and ZCTAs

Census tract

A small, stable statistical area, usually 1,200 to 8,000 people.

Nests inside a county, the way counties nest inside a state.

Most NaNDA measures are available at this level.

ZCTA (ZIP Code Tabulation Area)

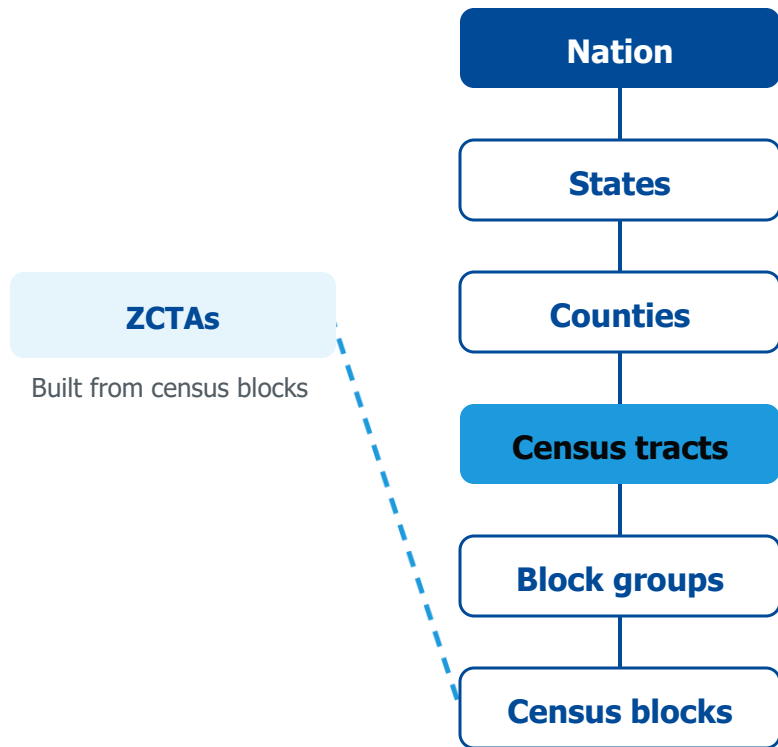
The Census Bureau's area version of a ZIP code, built from census blocks.

Not the boundary the Postal Service uses, and not nested inside anything.

ZIP ≠ ZCTA

A ZIP code is a mail delivery route. A ZCTA is a statistical area. They are **not** interchangeable. ZIP codes need a crosswalk to ZCTAs (Intermediate Session).

The Census geography hierarchy



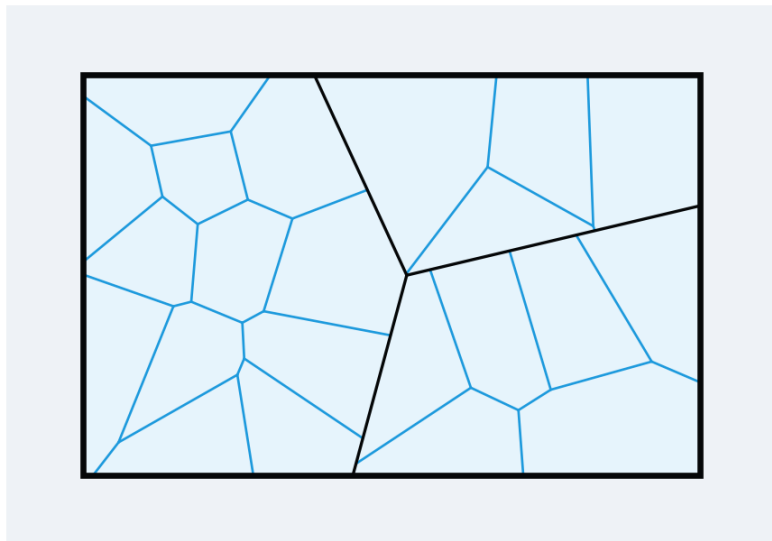
The center spine nests. Nation, states, counties, tracts, block groups, blocks. Each level fits inside the one above it.

ZCTAs sit off to the side. They skip tracts and counties entirely, which is why their lines never match tract or county lines.

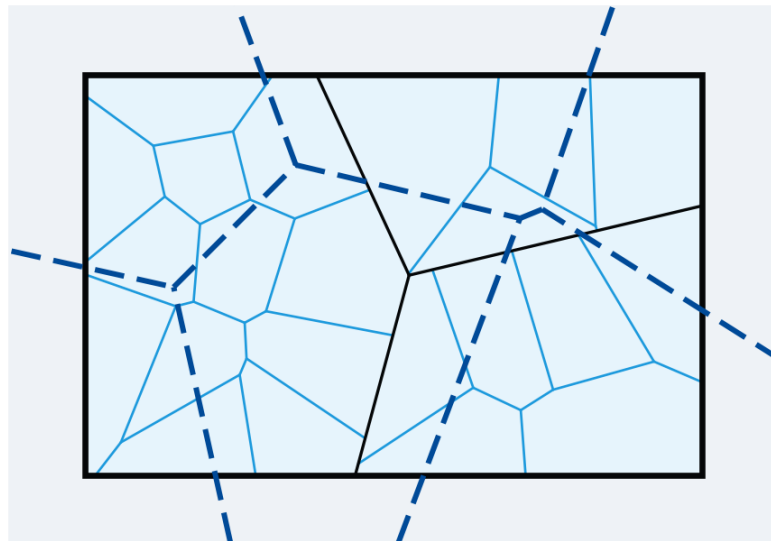
Simplified from the U.S. Census Bureau hierarchy diagram

Tracts and ZCTAs

Tracts nest in counties,
counties nest in the state



ZCTAs cross tract, county,
and state lines



How a FIPS code grows

State

2 digits

2 6

Michigan

County

5 digits

2 6 1 6 1

State, then county 161 (Washtenaw)

Census tract

11 digits

2 6 1 6 1 4 0 0 1 0 0

County, then tract 400100

Block group

12 digits

2 6 1 6 1 4 0 0 1 0 0 1

Tract, then block group 1

NaNDA's tract identifiers, **tract_fips10** and **tract_fips20**, are this 11-digit code. Stored as text to preserve leading zeros.

[Census Bureau: Understanding Geographic Identifiers](#)

ZCTA codes are a separate system

Census tract FIPS code

11 digits · state + county + tract

2 6 1 6 1 4 0 0 1 0 0

■ State ■ County ■ Tract

Nested. Every digit tells you where the tract sits.

ZCTA code

5 digits · no state or county inside it

4 8 1 0 4

■ ZCTA

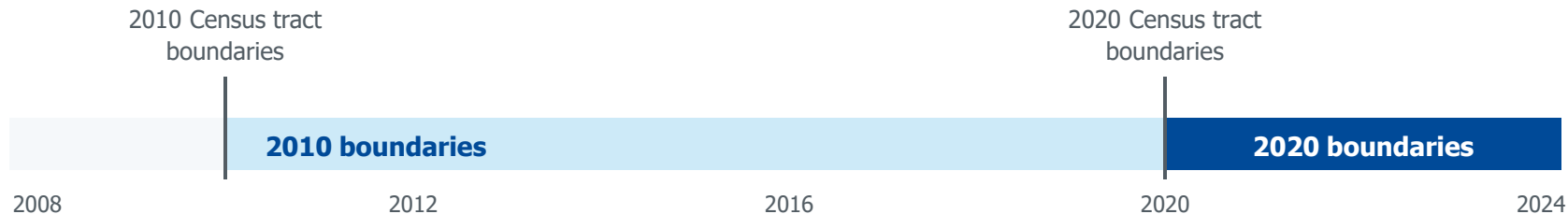
Not nested. It looks like a ZIP code, and usually matches one, but it is a Census area, not a mail route.

Have ZIP codes? They still need a crosswalk to ZCTAs before linking to NaNDA.

Geographic boundaries

Tracts and ZCTAs are redrawn after each Decennial Census. 2010 and 2020 codes do not align.

Match the boundary to when your geography was assigned, not to the survey year.



Your turn Jordan's addresses were geocoded in **2016**
Which geographic boundaries do Jordan's tract codes use?

POLL 1 OF 3 · BOUNDARIES

Which geographic boundaries fit Jordan's research?

Civic, Social, and Religious Organizations comes in four files. Jordan's data: **census tracts** on **2010 boundaries**.

A. Census Tract 2010

B. Census Tract 2020

C. ZCTA 2010

D. ZCTA 2020

E. Not sure

Join the poll Watch for it to pop up in your Zoom window.

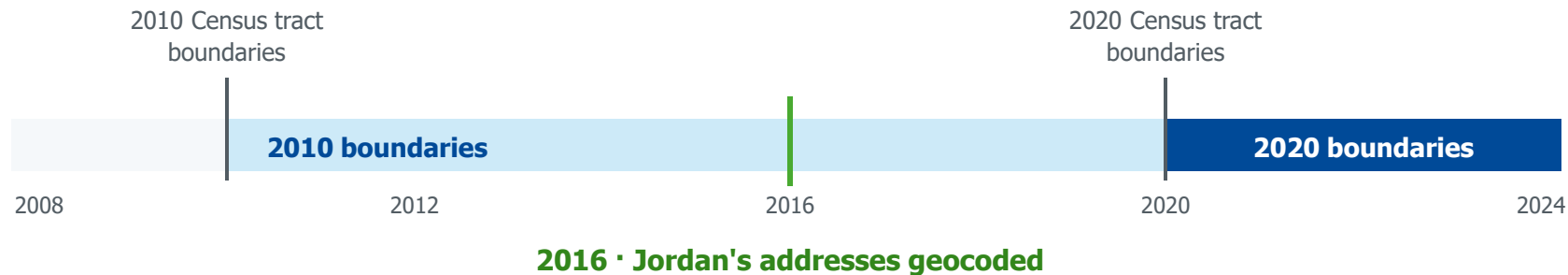
The answer is A: Census Tract 2010

		UNIT	BOUNDARIES
A	Census Tract 2010. Tracts, on the same geographic boundaries as Jordan's codes.	Match	Match
B	Census Tract 2020. Right unit, but the codes are different. It won't link to Jordan's 2010 tract codes.	Match	Wrong boundaries
C	ZCTA 2010. Right boundaries, wrong unit. Jordan has tract codes, not ZCTAs.	Wrong unit	Match
D	ZCTA 2020. Neither.	Wrong unit	Wrong boundaries

Boundaries first. The filename tells you the unit and the geographic boundaries. Only the Census Tract 2010 file links on **tract_fips10**. Years come next.

Jordan's boundaries: 2010

Geocoded in 2016, five years before the 2020 boundaries existed. Anything geocoded before 2020 almost certainly uses 2010 boundaries.



Jordan's Where

2010 boundaries. Use the **Census Tract 2010** file, linked on **tract_fips10**. Not sure when your data were geocoded? Ask whoever did it, or check the codebook.

NaNDA location variables

VARIABLE

tract_fips10

tract_fips20

↳ **tract_fips22**

zcta10

zcta20

WHAT IT IS

Census tract, 2010 boundaries (11 digits)

Census tract, 2020 boundaries (11 digits)

The same 2020 tracts, recoded for Connecticut's 2022 boundary change. An additional variable in the 2020 files, not a separate dataset. Use it if your data were geocoded after 2022 and include Connecticut.

ZCTA, 2010 boundaries (5 digits)

ZCTA, 2020 boundaries (5 digits)

The suffix tells you the geographic boundaries. Recent NaNDA datasets use these variable names; older ones may differ, so check the documentation.

TRY IT

Search NaNDA for 2010 census tract datasets



1 Open the [NaNDA repository](https://icpsr.umich.edu/sites/nanda) on ICPSR.

icpsr.umich.edu/sites/nanda

- Find Data
- Studies

2 Search "census tract 2010"

- topic for Jordan's research: "religious", "civic", or "social"

3 Open any study.

- Select "Data & documentation"
- Find the Census Tract 2010 file

4 Post the study title in the chat.

WHAT YOU WILL SEE

Most NaNDA studies come as the same four files.

Census Tract 2010

Census Tract 2020

ZCTA 2010

ZCTA 2020

Jordan's data fit the first one. Yours may fit a different one.

Where, so far

Where?

- Tracts nest in counties. ZCTAs nest in nothing.
- ZIP $\neq$ ZCTA. ZIP codes need a crosswalk first.
- Boundaries follow the geocoding date: before 2020 means 2010.
- Store FIPS codes as text to preserve leading zeros.

JORDAN'S WHERE

Census tract codes on 2010 boundaries. Census Tract 2010 file, linked on [tract_fips10](#).

Want a refresher after today? [NaNDA's guide to geographies](#)

SEGMENT 2

When?

Which NaNDA years match my data collection window?



NaNDA

Two kinds of timing



Yearly dataset

Civic, Social, and Religious
Organizations

one value per place per year

Period dataset

Socioeconomic Status

2008 to 2012

2013 to 2017

2016 to 2020

2018 to 2022



YOUR DATA

Jordan's interviews

2016 2017 2018 2019

2008

2011

2014

2017

2020

2023



WHEN, PART 1

Yearly dataset

Places to gather come from **Civic, Social, and Religious Organizations**: one value per tract per year.



NaNDA

Which years does Jordan use?

Civic, Social, and Religious Organizations is a yearly dataset, 1990 to 2022. Jordan's interviews: **2016, 2017, 2018, 2019.**

A. 2016 only, the enrollment year

B. 2016, 2017, 2018, and 2019, one per interview year

C. 2022 only, the most recent year

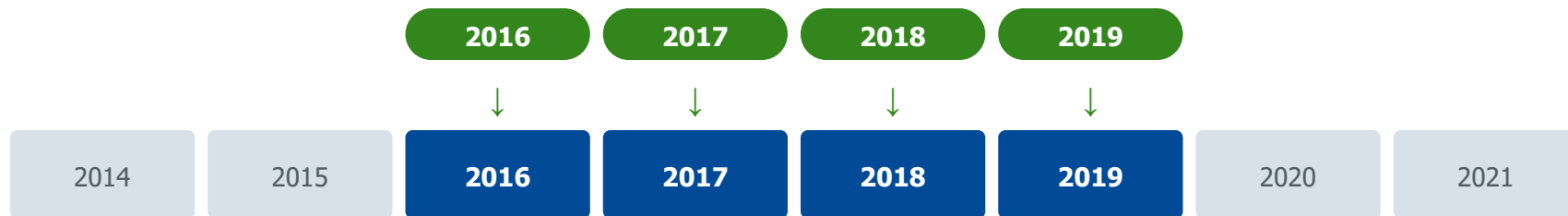
D. Not sure

Join the poll

Watch for it to pop up in your Zoom window.

The answer is B: every interview year

Jordan's interviews



Census Tract 2010 file • one value per tract per year, 1990 to 2022 (2014 to 2021 shown)

A yearly dataset has a value for every tract in every year, so each interview is matched to its own year. A 2016 interview gets the 2016 value for that tract; a 2019 interview gets the 2019 value.

Yearly:
match the year.

Every one of Jordan's years has a match. If your years fall outside the dataset's range, use the nearest year and document the decision in your methods.

One year, or every year?

Whether you need to match every interview year depends on how fast the neighborhood measure changes. That is a methods decision, and yours to make.

CHANGES SLOWLY

One year can be enough

Places of worship and civic organizations do not move around much. Using one year for everyone, say 2016 at enrollment, may work for Jordan, and keep the analysis simpler.

2016

2017

2018

2019

CHANGES FAST

Match each year

Some measures shift rapidly within a study window. Business openings and closings before and after the pandemic, for example. Match the interview year.

2016

2017

2018

2019

**Either way,
write it down.**

State which years you used and why in your methods. Not sure how fast your measure changes? Pull two years and compare them.

WHEN, PART 2

Period dataset

Neighborhood poverty comes from a **period dataset**: one value per 5-year window, not one per year.



NaNDA

Socioeconomic Status: the files

2010 tract boundaries
2020 tract boundaries



Boundaries first. Jordan's codes are 2010 tracts, so we'll use **Census Tract 2010** files. That leaves one question: **which time period?**

Which 5-year window fits Jordan's research?

Jordan's codes are 2010 tracts. Socioeconomic Status windows: **2008 to 2012** and **2013 to 2017** (2010 boundaries); **2016 to 2020** and **2018 to 2022** (2020 boundaries). Interviews: 2016 to 2019.

A. 2013 to 2017

B. 2018 to 2022

C. 2008 to 2012

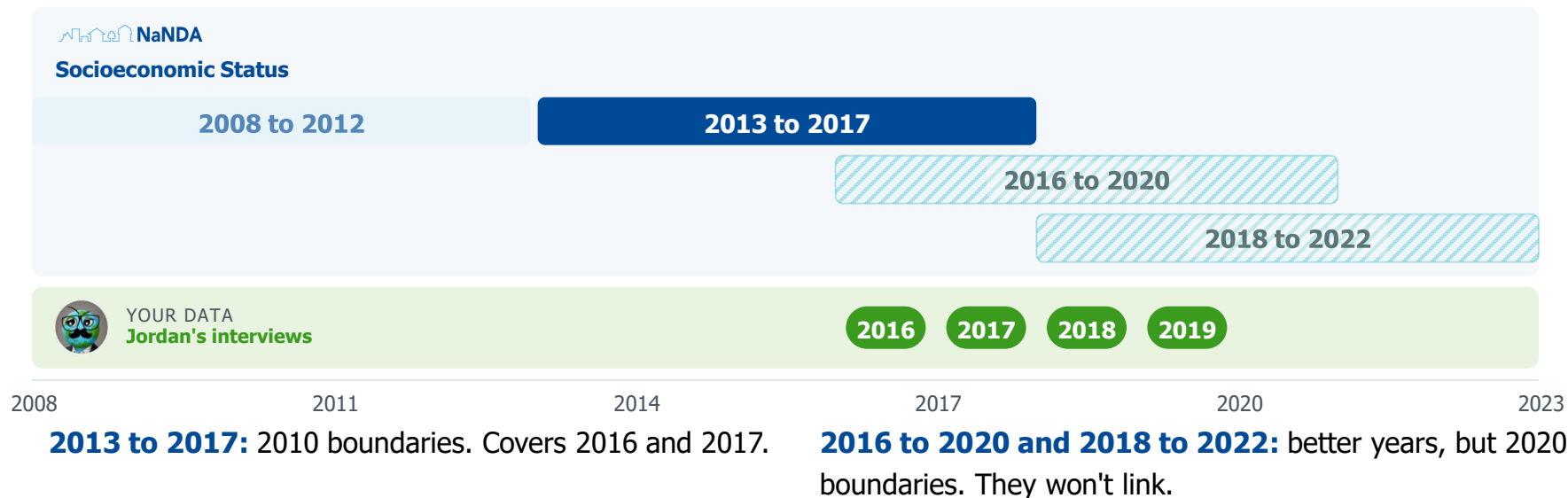
D. 2016 to 2020

E. Not sure

Join the poll Watch for it to pop up in your Zoom window.

The answer is A: 2013 to 2017

2010 tract boundaries
2020 tract boundaries



Boundaries first.

Note the gap for 2018 and 2019 in your methods.

What if Jordan's interviews ran 2011 to 2014?

2010 tract boundaries



2008 to 2012 overlaps 2011 and 2012. Closest to enrollment, when the addresses were collected.

2013 to 2017 overlaps 2013 and 2014. Closest to the later interviews.

**Pick one
or use both.**

Both windows use 2010 boundaries, so either one links. Say which you chose and why in your methods.

Period dataset: neighborhood poverty

- 1 **Search** the [NaNDA repository](#) on ICPSR for "*poverty*".
- 2 **Title.** Topic, geography, and years.
- 3 **Open Data & documentation.** Filenames include geographic boundaries and years.
- 4 **Check the data dictionary.** Preview variable names and labels (e.g., tract_fips10, Census Tract FIPS code, 2010) * must be logged in to ICPSR to preview
- 5 **Save the citation.** Add the dataset DOI (or full citation) to your notes or citation management software.

ICPSR STUDY PAGE

Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 1990-2022

Census Tract 2010

Census Tract 2020

ZCTA 2010

ZCTA 2020

tract_fips10	Census tract, 2010 boundaries
year	5-year ACS window, e.g. 2013 to 2017
ppov13_17	Proportion of people below the poverty line

Save Dataset DOI (digital object identifier)

Where and When, so far

Where?

- Census tracts nest within counties. ZCTAs nest in nothing.
- ZIP $\neq$ ZCTA. ZIP codes need a crosswalk first.
- Boundaries follow the geocoding date: before 2020 means 2010.

Jordan: 2010 tracts, linked on tract_fips10

When?

- Yearly datasets: match the interview year.
- Period datasets: pick the window that overlaps most, and say so.
- Boundaries first, then years. A year or two of gap is fine.

Jordan: Civic 2016 to 2019, every year. SES 2013 to 2017, gap noted in methods.

Want a refresher after today? [NaNDA's guide to geographies](#)



SEGMENT 3

What?

Find a dataset yourself: a five-minute scavenger hunt.



NaNDA

Scavenger hunt

Pick one prompt. Search NaNDA on ICPSR, find a dataset that fits, and post its name in the chat.



1

Food access: grocery stores

Years 2016

Geography Census tract, 2010
boundaries

2

Getting around: public transit stops

Years 2016 to 2018

Geography ZCTA

3

Staying connected: Internet access at home

Years 2015 to 2019

Geography Census tract

Tip: search a single plain word, like *grocery*, *transit*, or *internet*.

SEGMENT 4

Your turn

Find one dataset you would actually consider using for your own research question.



NaNDA

How the next 8 minutes run

0 to 4 min

Work on your own

Fill in Where, When and What for your project. Browse topics on the NaNDA website, then open the dataset on ICPSR for its files and variables.

4 to 7 min

Share

In the chat or unmute: dataset, geography, years, and why it fits.

7 to 8 min

Questions

Anything about finding, downloading, or linking a dataset.

Stuck?

Not sure what location data you have? Check your codebook or data use agreement.

Have ZIP codes? Crosswalk to ZCTAs first.

Can't find your topic? Try browsing the [NaNDA data page](#). Still stuck? Email nanda-admin@umich.edu, or [book Office Hours](#).

WRAP-UP

Keep your handout

When we reconvene, share one takeaway with the full group. The handout stays open after today, and the scavenger hunt answers are at the bottom.



NaNDA